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The effect of macroeconomic variables on GDP and the fiscal policy

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Abstract
The main objective of this paper is to determine which factors influence GDP growth and what are their effects with the particular focus on the fiscal policy. While most papers use structural models to estimate these effects I take the empirical approach.By using VECM I estimate impulse response functions to determine the biggest factors driving GDP growth are consumption and investment closely followed by government spending. While, unemployment lowers GDP growth significantly.

Keywords
GDP growth — Fiscal policy — VECM

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Introduction

During the last century macroeconomists were focused on one fundamental question. What drives economic growth? This problem is first tackled in Solow (1956). The author shows that an increase in savings leads to higher output steady state. However, long-term economic growth can only be sustained by an increase in productivity. This model, while providing good intuition for long-term growth has one important drawback it assumes a constant exogenous savings rate. To fix this Peter et al. (1965) uses an endogenous savings rate that depends on utility maximization of the consumers. In essence, consumers try to decide the best trade-off between consuming today and consumption in retirement. This new dynamic of youth and retirement introduces generational interactions within the model. The results of the paper further reiterate that economic growth can only be achieved by technological progress. This is because if you save more, thus buy more capital, it has diminishing returns on your output so you eventually reach a steady state. Conversely, Romer (1990) argues that investment in research and development drives long-term economic growth by creating technological progress. He suggest that new ideas are non-rival , which means that multiple people can use it at the same time,which is not applicable to forms of physical capital. This non-rivalry is what prevents returns from R&D from stopping, which is why government should subsidize research.

Besides the long-term economic growth the influence of other factors are explored due to their impact on the short-term economy and policy. One of the most looked at statistics is the fiscal multiplier. The core idea behind the fiscal multiplier is that if we increase government spending by 1 dollar by how much will GDP increase? Ilzetzki et al. (2013) uses SVAR impulse response function to compute fiscal multipliers for developed and developing economies. He finds that for developed economies the fiscal multiplier is 0.39 meaning that 1 dollar increase in government spending increases GDP by 39 cents. However, for countries with lower development level fiscal multiplier is not significantly different from 0. Another well known relationship is between unemployment and GDP commonly known as Okun's law. Ball et al. (2017) reexamines Okun's law in different world countries (E.g Spain,Japan, United Kingdom). He finds that the relationship between unemployment and GDP is negative and significant for all regression specifications.

The aim of my research is to reexamine the impact of previous literature using less used empirical methods. Mainly, I analyze the effect of labor productivity, investment, fiscal policy,inflation and unemployment on GDP, while adding new rarely explored variable such as federal funds rate.

1. Data

For my research I use FRED time series of different macroeconomic factors. From Fred data I import labor productivity,government spending, GDP , consumption, investment, inflation and federal funds rate quarterly data from 1984 to 2020.

The cutoff at 2020 is chosen because of covid-19 pan-

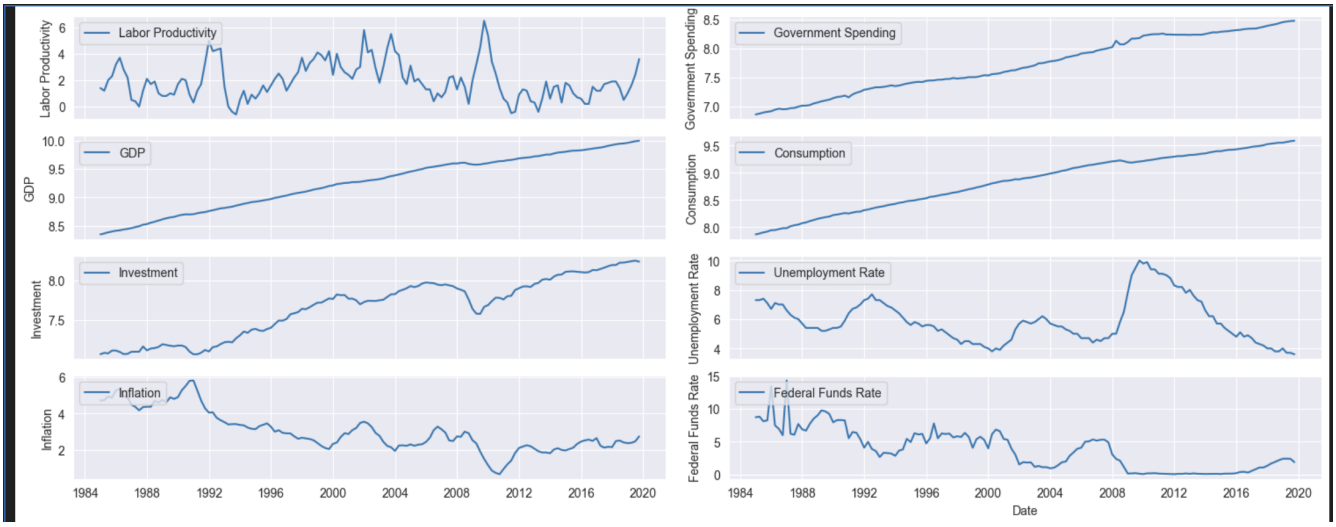

Figure 1. All series over time

demic that is problematic due to a huge surge in unemployment and a massive increase in inflation, which makes this period very troublesome to deal with.

2. Explanatory data analysis

I start my analysis by examining the main data trends. Firstly, time series is plotted over time to inspect trend and seasonality. Figure 1 shows that most of the series such as GDP, consumption, government spending exhibit a strong linear trend. While, labor productivity and federal funds rate seem to be more stable. To check the unit root formally I run augmented dickey-fuller test for all variables presented in Figure 1.

For variables that have a linear trend I use dickey fuller test with the trend component. Dickey fuller test reveals that all variables are insignificant at 5% confidence level, meaning that we cannot reject the null hypothesis that they have a unit root. After differencing I rerun the dickey fuller test this time using constant and no trend. After data transformation all variables become stationary as indicated by p-value of dickey fuller test being less than 0.05. This shows that the variables have integration order I(1).

Furthermore, I check what is the optimal lag for the model. I do this by evaluating AIC, BIC, FPE and HQIC. BIC and HQIC indicate the lag of 0, while AIC thinks that a lag of 10 is optimal. I follow BIC and choose lag order of 1 because BIC heavily penalizes overfitting Besides this, I also look at Johansen test for cointegration (that linear combination of my variables is stationary). The test results in cointegration rank of 2.

Last part of the explanatory data analysis is correlation analysis. Firstly, I look at the ACF's and PACF's for every single differenced variable to double check for seasonality. If the data is seasonal there should be a a spike in correlation for example every 4 quarters. Luckily, most of the auto correlations are insignificant indicating no seasonality. Later I check the correlation matrix between the variables. The most notable correlation is between unemployment and consumption , investment and GDP being at around -0.4 , which is consistent with Okun's law.

3. Data Preprocessing

During the data preprocessing part I conduct feature engineering. Firstly, I take the logarithm of the nominal variables such as GDP or investment because I want to put the variables in a similar scale. This is also done so that parameters of the models do not explode and I can model the variables in terms of percentages. Secondly, I use Cholesky decomposition to othognalize the shocks and model wherever the variables affect other variables contemporaneously. I put the variables in this order: Labor productivity, government spending,GDP, consumption, investment, unemployment rate, inflation, federal funds rate. I place labor productivity first and government spending second because they are assumed to be exogenous in most economic models. Then I put GDP third because I want other macroeconomic variables to affect GDP with a lag. The other variables that come after GDP affect GDP with their corresponding placement after GDP for example consumption affects GDP with a lag of 1 and investment affects GDP with a lag of 2. Finally, there comes policy shocks such as interest rates that take a long time to affect GDP. The last change I make to model the variables is that I create a couple of dummy variables to control for periods of high economic intensity. One of the periods is in 1986-1987 where there were abnormaly high interest rate hikes reaching up to 15%. Secondly I control for the 2008 crisis specifically the months of July and October because it's the period where Lehman brothers collapsed and unemployment surged to more than 10%.

4. Methodology

Explanatory data analysis indicates that the variables are cointegrated. This means that SVECM outperforms SVAR because it models short-term dynamics in differences and accounts for long-run dynamics by including long-term cointegration adjustment . Furthermore, since SVECM uses differences, it is important that my variables are I(1), which my data satisfies as seen previously in the dickey fuller tests. Since I use logarithms in the data preprocessing for nominal variables after differencing my variable become in terms of log differences, which can be interpreted as growth rate. Thus, we can interpret impulse response functions (IRF) as how much would GDP change in percent if there was a 1 standard deviation shock in the variable.

$$A\Delta y_t = \alpha\beta' y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i^* \Delta y_{t-i} + \Phi D_t + \varepsilon_t \tag{1}$$

- A: 8×8 structural matrix defining contemporaneous impacts.
- Δy_t: Vector of first-differenced endogenous variables (log-transformed).
- $\alpha\beta' y_{t-1}$: The Error Correction Term (ECT), where β' represents the cointegrating vectors and α represents the adjustment speeds.
- Γ_i^*: 8×8 matrices of short-run autoregressive coefficients at lag i.
- ΦD_t: Matrix of coefficients for exogenous dummy variables (structural breaks)
- ε_t: Vector of structural shocks, assumed to be i.i.d. $N(0, I)$.

From this model we can iteratively unwrap the model to get the a representation of the model in terms of current and previous periods shocks to different variables and calculate derivatives with respect to those shocks to know what is the effect of a variable on GDP growth.

5. Results

After estimating the model using maximum likelihood estimation I draw the impulse response functions.

Figure 2 shows the effect of labour productivity, government spending, GDP and consumption on GDP. From the graph it can be seen that 1 standard deviation increase in labour productivity only slightly increases GDP growth. However this effect is insignificant because 0 falls inside the confidence interval. Interestingly, 1 standard deviation increase in government spending increases GDP by around 0.5% in the long run confirming that fiscal policy is not strong enough to stimulate economy. Lastly, 1 standard deviation increase in consumption increases GDP by 1%

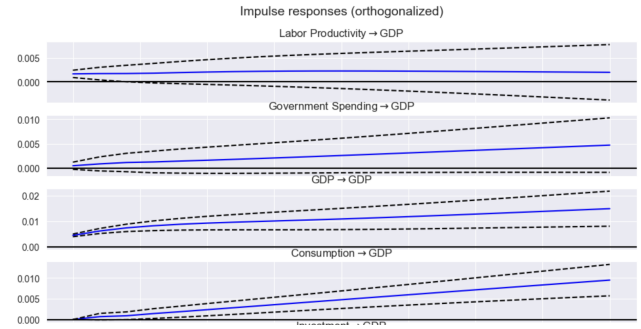

Figure 2. Impulse response functions for the first 4 variables

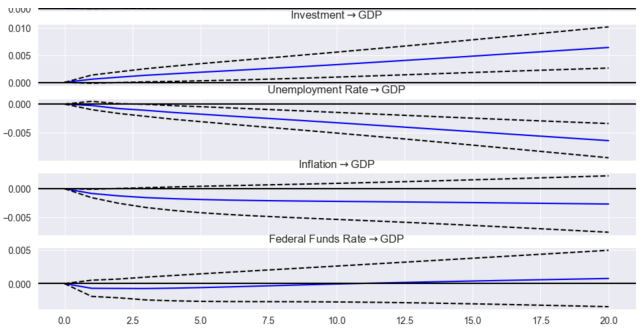

Figure 3. Impulse response functions for the last 4 variables

Figure 3 shows the effect of investment, unemployment, inflation and federal funds rate. From the graph we can see that 1 standard deviation increase in investment increases GDP by around 0.7%, while 1 standard deviation increase in unemployment decreases GDP by around 0.5% in the long-term, which are both highly statistically significant. Inflation also decreases GDP growth. Unfortunately,the confidence intervals of this effect cross the x axis making it insignificant. Lastly, we can see that federal funds rate have no lasting affect on GDP growth.

6. Conclusions

The study empirically estimates the effect of different economic variables on GDP growth. The data suggests that one of the main drivers of GDP is consumption and investment. While, unemployment rate puts significant downward pressure on GDP. I also further reexamine the affect of fiscal policy on GDP, while fiscal policy increases GDP growth in both short and long-runs, the estimated effect is not high enough to prove that fiscal multiplier is more than 1.

7. AI Usage

- AI was used for debugging and code readability.
- AI was used for making stylistic changes to the LaTeX file.
- AI was used for making little grammatical improvements in the text.

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